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Prioritization of patients at risk of heart attack using a novel full-objective ITARA based on Random Forest and Decision tree

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Heart attacks remain a major cause of morbidity and mortality, particularly among middle-aged and older adults, often aggravated by unhealthy lifestyles and limited preventive care. Early identification and prioritization of at-risk individuals are essential to avoid severe complications. While prediction models exist, they often lack robust frameworks for prioritization across multiple clinical factors. This study addresses the gap by adapting Indifference Threshold Based Attribute Ratio Analysis (ITARA), a multi-criteria decision-making (MCDM) method, into a fully objective framework enhanced by machine learning. Unlike traditional ITARA, which relies on expert-defined thresholds, the proposed approach derives thresholds from variable importance scores generated by classification models. Among the models tested, Random Forest achieved 97% accuracy and was used to produce feature importance values. These scores refined ITARA, improving prioritization accuracy. Results identified the number of major vessels (Ca) as the most critical predictor. By integrating machine learning, the ITARA framework becomes a transparent, data-driven tool that improveWs both early detection and the clinical management of heart attack risks.

Keywords Heart attack, Multiple criteria decision making, Indifference Threshold-based Attribute Ratio Analysis (ITARA), Random Forest, Decision tree, Classification, Machine learning

Cardiovascular diseases (CVDs) are among the leading causes of mortality worldwide, encompassing a range of conditions such as coronary heart disease, cerebrovascular disease, and rheumatic heart disease. These disorders can compromise the heart's ability to pump sufficient blood to sustain normal physiological functions, often resulting in myocardial infarction (MI), commonly known as a heart attack[1]. MI typically manifests with symptoms such as chest pain radiating to the left arm or neck, shortness of breath, sweating, nausea, fatigue, and irregular heart rhythms[2]. Despite advances in prevention and treatment, CVDs continue to pose a significant public health challenge, influenced by behavioral, physiological, and genetic risk factors.

Behavioral risk factors, including unhealthy diet, physical inactivity, tobacco use, and excessive alcohol consumption, contribute to physiological changes such as elevated blood pressure, high blood glucose levels, and lipid abnormalities. These risks are further compounded by obesity and diabetes, creating a high-risk profile for cardiovascular complications[3]. Genetic predisposition, age, and gender also play important roles, with older adults and males being particularly vulnerable. Although lifestyle modifications and pharmacological interventions, including ACE inhibitors, beta blockers, and statins, can help mitigate these risks, the global burden of CVDs continues to rise[4].

The World Health Organization estimates that approximately 17.9 million people die each year from cardiovascular diseases (CVDs), with more than four-fifths of these deaths attributed to heart attacks and strokes. Alarmingly, one-third of these deaths occur prematurely in individuals under the age of 70, highlighting the urgent need for timely diagnosis and intervention[5]. Early recognition of symptoms, such as chest pain or shortness of breath, is essential for initiating prompt treatment and preventing severe complications. However, many patients encounter barriers to accessing timely care, including limited healthcare resources, lack of awareness, and delayed symptom recognition.

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To address these challenges, advancements in data-driven methodologies, particularly machine learning, offer promising solutions. Techniques such as decision trees and random forests have demonstrated high accuracy in predicting heart attack risk by analyzing large datasets and identifying complex patterns among risk factors. These methods provide objective and reproducible insights into the likelihood of cardiovascular events, enabling healthcare providers to identify high-risk individuals with greater precision. In particular, random forest models, with their ensemble learning capabilities, often outperform traditional predictive approaches by effectively capturing interactions among variables and determining their relative importance.

Given the global scale of cardiovascular disease, prioritizing high-risk individuals for early intervention is essential for reducing CVD-related mortality. Traditional risk assessment models often focus on predicting the likelihood of heart attacks but lack robust frameworks for prioritizing patients for timely intervention. This limitation is especially critical in resource-limited settings, where healthcare systems must allocate resources efficiently.

Although machine learning excels at risk prediction, translating these predictions into actionable patient prioritization requires additional methodological development. Multi-Criteria Decision-Making (MCDM) frameworks, such as ITARA, provide a structured approach to ranking individuals across multiple criteria. However, they traditionally rely on expert-defined indifference thresholds, which can introduce subjectivity, variability, and scalability limitations. Moreover, most existing MCDM methods have limited integration with data-driven feature weighting.

To address these challenges, we propose a novel hybrid framework that integrates classification-based feature importance into ITARA. Unlike classical methods such as AHP and TOPSIS, which depend on subjective judgments and do not account for the underlying structure or distribution of the data, this integration creates a fully data-driven, robust, and reproducible prioritization process. This represents a unique methodological contribution, making the framework objective, scalable, and directly applicable to healthcare analytics.

In this hybrid approach, variable importance scores derived from the best-fit classification model are used to define thresholds for each risk factor, thereby transforming ITARA into a data-driven method. This process identifies the most critical risk factors contributing to cardiovascular mortality by calculating their respective weights within the ITARA framework. Risk factors with higher weights are recognized as key determinants for guiding patient prioritization, ensuring that healthcare resources are allocated to those at greatest risk. Ultimately, this enables more accurate and personalized patient management.

The machine learning enhanced ITARA framework bridges the gap between risk prediction and actionable prioritization. By leveraging data-driven insights within a Multi-Criteria Decision-Making process, the model provides accurate early detection and a systematic method for managing heart attack risk. Translating predictive analytics into clear priorities enables more personalized patient care, optimizes resource allocation, and has significant potential to reduce the global burden of cardiovascular diseases.

The paper is organized as follows: Sect. "Related work" reviews relevant terminology and related work, Sect. "Materials and methodology" details the proposed methodology, Sect. "Result and discussion" presents the results, Sect. "Managerial implications" discusses the managerial implications, Sect. "Conclusion" concludes with a summary of the key contributions, and Sect. "Future work" outlines directions for future research.

Related work

Heart disease includes a variety of conditions that affect the heart's structure and function. Many cases result from blocked or damaged coronary arteries, which reduce oxygen supply and can lead to chest pain, myocardial infarction (MI), or heart failure. Irregular heart rhythms and valve disorders can also impair cardiac function, affecting the entire body[6]. Timely and precise diagnosis is essential to prevent complications and reduce mortality. However, nonspecific symptoms such as fatigue, dizziness, nausea, and shortness of breath often complicate the diagnostic process[7].

Traditional diagnostic methods involve evaluating medical history, risk factors (e.g., age, coronary artery disease, diabetes, smoking), physical examinations, blood tests, and electrocardiograms (ECG), all of which support accurate diagnosis and treatment planning[8]. Advancements in computational methods, particularly machine learning (ML), have significantly enhanced heart disease diagnosis[9]. ML techniques analyze complex medical data, providing precise predictions that support early diagnosis and intervention[10].

The proliferation of digital health technologies and widespread monitoring devices has made large-scale medical data readily available for analysis. Models trained on these datasets can support Clinical Decision Support Systems (CDSS), assisting physicians in making faster and more accurate diagnoses. These systems are also valuable for training non-specialists, thereby improving overall healthcare decision-making and patient outcomes[11].

Numerous studies have highlighted the potential of machine learning models for heart disease diagnosis, emphasizing not only predictive accuracy but also interpretability and clinical applicability. Chang et al.[12], compared Support Vector Machines (SVM), Random Forest, XGBoost, and k-Nearest Neighbors (k-NN), finding that SVM and XGBoost achieved the highest predictive accuracy. Using interpretability techniques such as SHAP and LIME, they identified key features, including the number of major vessels (CA), chest pain (CP), thalassemia (thal), maximum heart rate (thalach), and ST depression (oldpeak), as the most influential predictors. These tools provided clinicians with clear insights into how each feature contributed to classification outcomes, thereby supporting data-driven decision-making.

Similarly, MA et al.[13], analyzed the Cleveland Heart Disease dataset from the UCI Machine Learning Repository, achieving 86.89% accuracy with Logistic Regression and demonstrating the effectiveness of traditional models when combined with feature selection. Building on this, Rao et al.[14], applied the k-Nearest Neighbor algorithm with parameter weighting, which achieved a higher accuracy of 91.4% and outperformed both Random Forest and Decision tree models in early heart disease detection.

Other studies have extended machine learning applications to related conditions. For instance, Liaw et al.[15], evaluated Naïve Bayes, SVM, Decision tree, and Logistic Regression for arrhythmia prediction, with Logistic Regression achieving the highest accuracy (87%) due to its alignment with the data's statistical assumptions. In the context of heart failure, Lo et al.[16], compared Logistic Regression, Random Forest, SVM, Kernel-SVM, and LightGBM, reporting Random Forest as the most effective method with 96% accuracy and 99% AUC. Similarly, Wang et al.[17], proposed a Cloud-Random Forest (C-RF) model for assessing coronary heart disease (CHD) risk, which achieved 85% classification accuracy on the Framingham dataset.

Beyond predictive accuracy, some researchers have explored combining data-driven methods with clinical expertise. Ben-Assuli et al.[18], compared machine learning feature sets with those provided by human experts for predicting CHF readmissions and demonstrated that a human–machine collaborative model (AUC 0.836) outperformed both approaches alone. Likewise, Qu et al.[19], emphasized the predictive value of laboratory-based clinical data over questionnaires, demonstrating that machine learning can identify maternal biomarkers associated with congenital heart disease risk.

Efforts have also been made to integrate machine learning into broader frameworks. Alsheikh et al.[20], developed a machine learning-based heart disease prediction model (ML-HDPM) that achieved strong and balanced performance across key metrics, highlighting its potential to support early diagnosis and improve patient care. Extending this line of work, Salehi and Khedmati[21], introduced a comprehensive framework that combines machine learning with multi-attribute decision-making techniques to predict CHD. Their model incorporated lifestyle, maternal, environmental, nutritional, and socioeconomic factors, enabling personalized risk assessment through patient clustering.

Model comparison studies further reinforce these findings. Rimal et al.[22], applied cross-validation with advanced preprocessing to evaluate Logistic Regression, SVM, KNN, and Random Forest for heart disease prediction. While Logistic Regression and KNN achieved the highest single-run accuracy, Random Forest demonstrated superior overall performance. A result consistent with Shanmugapriya et al.[23], who reported its highest accuracy (87.85%) compared with other algorithms.

Despite these advances, most studies focus primarily on improving model accuracy, while relatively few address the critical challenge of prioritizing high-risk individuals identified by machine learning for timely intervention. This gap is particularly significant in resource-limited healthcare systems, where effective prioritization and resource allocation can substantially enhance patient outcomes.

Multi-Criteria Decision Making (MCDM) approaches, such as the ITARA (Indifference Threshold-Based Attribute Ratio Analysis) method, offer systematic tools for prioritization. ITARA combines objective analysis with expert judgment, using a predefined "indifference threshold" to assign weights to risk factors. While the objective aspect ensures consistency, the reliance on expert-defined thresholds introduces subjectivity, which can limit applicability[24].

This study integrates machine learning into ITARA, using models such as Random Forest and Decision tree to set data-driven feature thresholds, thereby reducing reliance on expert judgment and improving patient risk prioritization.

The MCDM process follows three steps: (1) identifying risk factors through literature review and expert input, (2) assigning data-driven weights, and (3) calculating risk scores to rank high-risk individuals. This integration of machine learning and MCDM enhances both accuracy and practicality, addressing key gaps in existing approaches.

In the following sections, the methodology employed in this research is discussed in detail, highlighting its innovative contribution to heart disease diagnosis and patient prioritization.

Materials and methodology

By combining machine learning with ITARA, this research predicts heart attack risk and prioritizes high-risk individuals to improve patient care. Figure 1 outlines the system framework, and the following subsections describe the methodology and materials.

Dataset

The dataset used in this study is a comprehensive heart disease prediction dataset, sourced from Kaggle, a widely recognized platform for open-access datasets[25]. It contains 1888 records, with 911 patients (48%) classified as having a low risk of heart attack and 977 patients (52%) classified as having a high risk. The dataset comprises 14 key features crucial for predicting heart attack and stroke risks, encompassing both medical and demographic factors, as outlined in Table 1.

A detailed description of each feature is provided below[26]:

1. *Age (age):* The risk of heart attacks increases sharply with age, making it a critical factor to consider in cardiovascular health.
2. *Sex (sex):* Research shows that men under 50 are at a higher risk of experiencing heart attacks than women. Following menopause, women's risk of heart attacks may become similar to men's, although this is still debated.
3. *Chest pain (cp):* This condition, known as angina, occurs when the heart muscle doesn't receive enough oxygen-rich blood. It is often accompanied by a sensation of tightness or intense pressure in the chest. Discomfort may also spread to the shoulders, jaw, back, or neck, and it can be associated with symptoms resembling indigestion. In some cases, the pain may extend to the hands.
4. *Blood pressure (trtbps):* High blood pressure can impact the arteries, often due to factors such as abnormal cholesterol levels, elevated blood sugar, obesity, and other conditions that increase the associated risks.

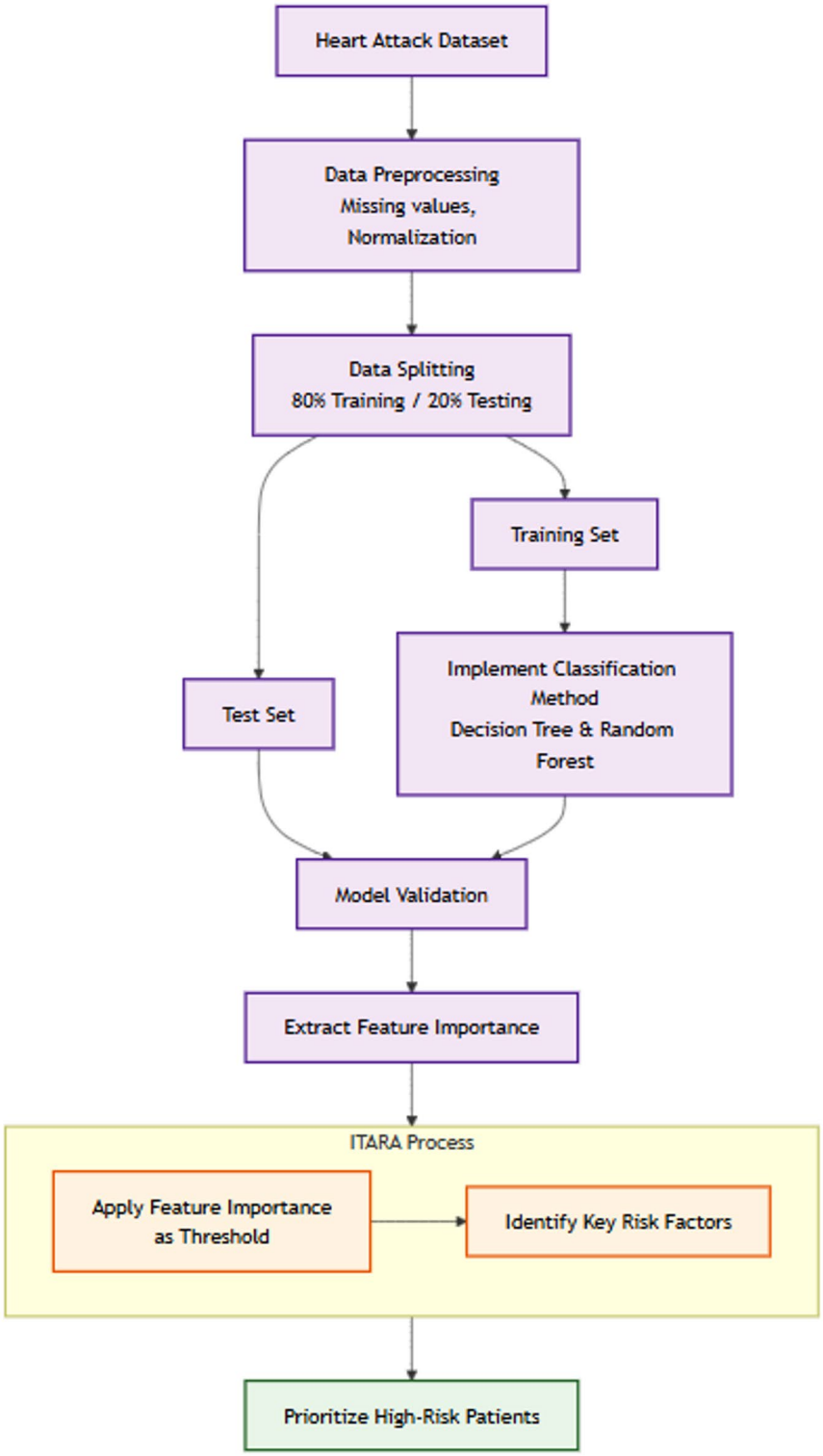


Fig. 1. The proposed system framework.

5. *Cholesterol (chol):* Imbalanced or harmful cholesterol can also affect the arteries by narrowing them, particularly low-density lipoprotein (LDL) cholesterol. High levels of blood fats, such as triglycerides, can further increase the risk of heart attacks. Therefore, maintaining healthy cholesterol levels is important for reducing the risk of heart attacks.
6. *Fasting blood sugar (fbs):* High blood sugar can contribute to the risk of a heart attack. This may occur due to reduced hormone production by the pancreas or a lack of response to insulin.

Features	Feature descriptions	Values
Age	Age of the patient	29–77
Sex	Gender of the patient	1 = Male, 0 = Female
Cp	Chest pain type	0 = Typical angina, 1 = Atypical angina, 2 = Nonanginal pain, 3 = Asymptomatic
Trestbps	Resting blood pressure	94–200 mm Hg
Chol	Serum cholesterol level	126–564 mg/dl
Fbs	Fasting blood sugar > 120 mg/dl	1 = true, 0 = false
Restecg	Resting electrocardiographic results	0 = Normal, 1 = ST-T wave abnormality, 2 = Left ventricular hypertrophy
Thalach	Maximum heart rate achieved	71–202
Exang	Exercise-induced angina	1 = yes, 0 = no
Oldpeak	ST depression induced by exercise relative to rest	0–6.2
Slope	Slope of the peak exercise ST segment	0 = Upsloping, 1 = Flat, 2 = Downsloping
Ca	Number of major vessels (0–3) colored by fluoroscopy	0–3
Thal	Thalassemia types	1 = Normal, 2 = Fixed defect, 3 = Reversible defect
Target	Outcome variable (heart attack risk)	1 = more chance of heart attack, 0 = less chance of heart attack

Table 1. Description of dataset.

7. *Resting electrocardiographic (resting):* In cases of moderate to high heart attack risk, the current situation does not provide enough insight into the drawbacks of screening. For individuals at lower risk, the potential negative effects of screening, such as skin irritation or rashes, can be outweighed by the benefits of exercise.
8. *Heart rate (thalach):* An increase in heart rate is linked to a higher risk of heart disease, similar to the rise in risk associated with elevated blood pressure. Research has shown that for every 10 bpm increase in heart rate, the likelihood of cardiac death rises by 20%.
9. *Angina (exng):* Exercise-induced angina causes discomfort that feels tight, gripping, or squeezing, ranging from mild to severe. The pain is typically felt in the center of the chest and may radiate to the shoulders, back, jaw, arms, or neck. Angina is an important indicator of coronary disease, making it valuable to treat it as a distinct category for analysis.
10. *Thalium stress test (thall):* The duration of the segment is crucial, as it's important to assess whether recovery is occurring consistently after peak stress in a positive treadmill test. Abnormal values are characterized by a downslope depression of 1 mm or less, lasting between 60 and 80 ms. Equivocal tests, such as those with up-sloping segments, can also occur during exercise.

Data preprocessing

In real-world scenarios, data collected from various sources often contain errors, missing values, or inconsistencies that can hinder machine learning model performance. To address these issues, the following preprocessing steps were applied[27]:

1. *Cleaning:* Missing values were removed to maintain data integrity.
2. *Encoding:* Categorical data, such as sex and chest pain type, were encoded into numerical formats suitable for machine learning algorithms.
3. *Scaling:* Numerical features like cholesterol and blood pressure were normalized to ensure uniformity, enhancing model convergence.

These preprocessing steps are also essential for ITARA, as they ensure that the input data is clean and well-structured for prioritization and multi-criteria decision-making analysis. In this study, the prepared dataset is used to train classification models. These supervised algorithms classify data into categories based on patterns learned from the training set. Proper preprocessing enhances model performance and increases the reliability of the proposed method. The dataset was cleaned by removing missing values and encoding categorical variables as numerical values, as summarized in Table 1.

Machine learning techniques

This section outlines the machine learning models used in this study to predict heart attack risk: Decision tree (DT) and Random Forest (RF). These algorithms were selected for their high accuracy, ability to capture complex non-linear relationships, and interpretability, which are crucial in healthcare data analysis. Critically, both models provide feature importance scores that highlight key risk factors, an essential component for this study, as these scores were used as the Indifference Threshold (IT) in the ITARA method, enabling objective, data-driven prioritization. While other models, such as SVM or neural networks, can achieve high predictive accuracy, they often lack interpretable and robust feature importance outputs, making them less suitable for integration with ITARA. By leveraging these importance measures, the models offer robust and actionable insights into patient risk prediction. The following subsection provides a concise overview of these algorithms.

Decision tree

A **Decision tree** is a classification and prediction model capable of handling both categorical and numerical data, frequently used in medical applications due to its intuitive and interpretable structure. It organizes information hierarchically into three key components:

1. *Root node:* Represents the initial decision point, encompassing the entire dataset.
2. *Internal nodes:* Represent intermediate decision splits based on selected features.
3. *Leaf nodes:* Provide final classifications or predictions based on the splits.

The model builds trees using algorithms like **CART (Classification and Regression Tree)** or **ID3 (Iterative Dichotomiser 3)**, employing criteria such as **Gini impurity** or **information gain** to split the dataset at each node[28].

Advantages

- *Simplicity:* Easy to implement and understand.
- *Interpretable results:* Provides clear, actionable insights.
- *Minimal preprocessing:* Handles categorical and numerical data directly.

Limitations

Overfitting: Prone to capturing noise in complex datasets.
Limited generalization: Single decision trees may not perform well on unseen data.

Decision trees organize information into a hierarchical, tree-like structure, making them intuitive and versatile, particularly for applications such as medical data analysis[29]. Their structure comprises three main components: the root node, which initiates decisions; internal nodes, representing intermediate decision points; and leaf nodes, providing the final classification or prediction[30]. This design simplifies complex datasets, enabling decision trees to be easily implemented and interpreted.

Decision trees are constructed using algorithms like CART (Classification and Regression Tree) or ID3 (Iterative Dichotomiser 3), which split data into subsets based on criteria such as Gini impurity or information gain. These methods ensure that meaningful patterns are captured within the tree structure. For example, in medical contexts, decision trees can analyze symptoms and test results to diagnose diseases[31].

Advantages of decision trees include their simplicity, visual clarity, and minimal preprocessing requirements. However, they also have limitations. Decision trees may overfit large or complex datasets, capturing noise instead of meaningful patterns. Techniques like pruning and ensemble methods, such as Random Forest, address these challenges by improving accuracy and generalizability.

To enhance accuracy in this study, we also implement the Random Forest model, which aggregates multiple decision trees to produce more robust and reliable predictions. Random Forest reduces the risk of overfitting by combining the predictions of individual trees, often outperforming a single decision tree. This ensemble learning approach not only leverages the strengths of decision trees but also minimizes their weaknesses.

The study further defines and explores the Random Forest model, delving into its functionality and advantages. By comparing the performance of Decision tree and Random Forest models, we evaluate the strengths and limitations of both techniques in handling the dataset.

Random Forest

Random Forests are ensemble learning methods that combine multiple decision trees to achieve better accuracy and reliability. They are particularly effective for classification tasks, such as heart attack risk prediction, due to their ability to capture non-linear relationships and feature interactions[32,33].

Key features

1. *Bagging (Bootstrap aggregating):* Each tree is trained on a random subset of the dataset, enhancing diversity and reducing overfitting.
2. *Majority voting:* For classification tasks, the final output is determined by the majority vote among all trees.
3. *Feature importance:* RF can rank features using metrics like Gini importance or mean decrease in impurity, helping identify the most influential predictors.
4. *Adjustable voting threshold:* Allows optimization of performance metrics like recall, precision, and F1-score.

Advantages

- *Higher accuracy:* Outperforms single decision trees by aggregating predictions.
- *Robustness:* Less sensitive to noisy data or overfitting.
- *Scalability:* Performance improves with an increasing number of trees.

Applications: Widely used in medical diagnostics, fraud detection, and image classification, RF models are known for their versatility and reliability. In this study, RF was chosen for its superior accuracy, stability, and robust feature importance measures, which enable clear, interpretable insights and support objective, data-driven prioritization, as illustrated in Fig. 2.

In the medical field, accuracy alone is an insufficient metric. Therefore, after training and assessing both the Decision tree and Random Forest models, their performance is compared using a variety of metrics to identify the best model, as detailed below.

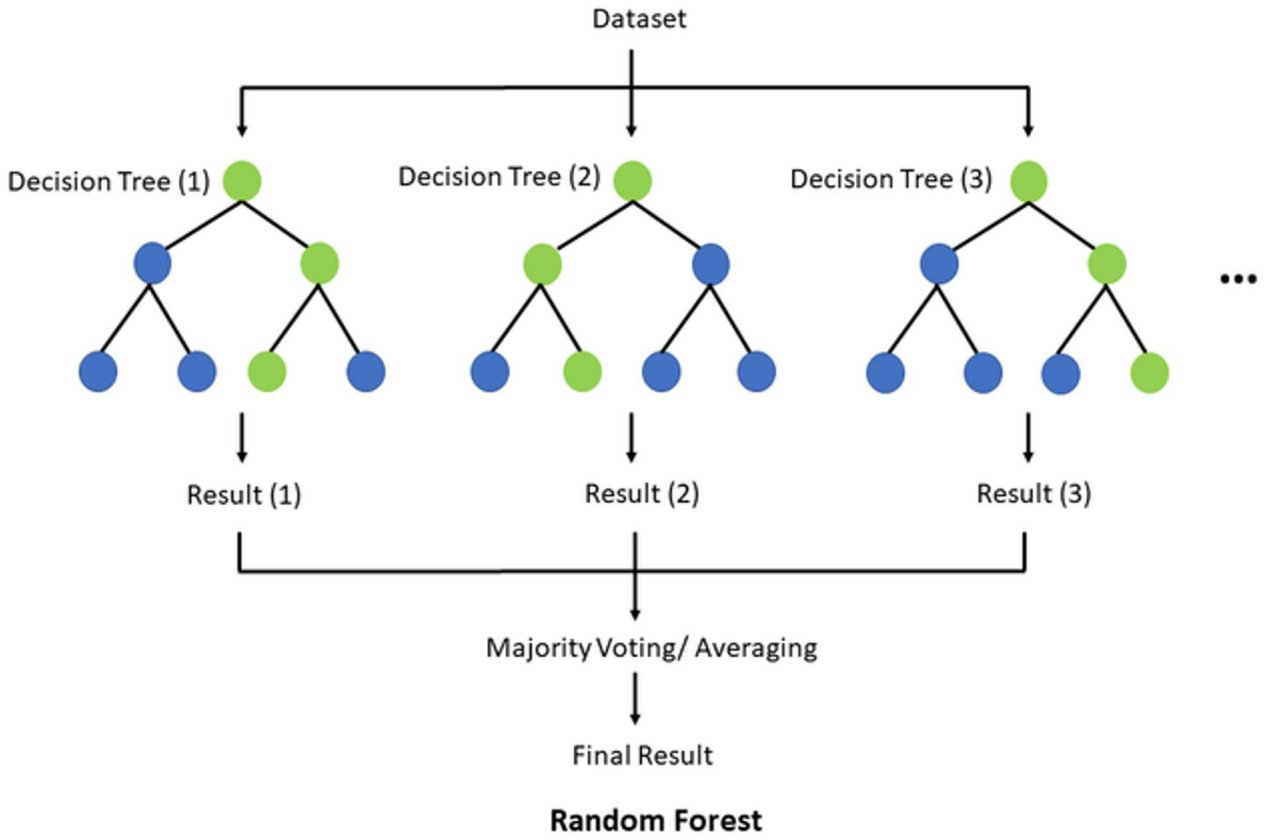


Fig. 2. RF model classification process.

Basic definitions

- *True positive (TP):* The number of individuals correctly identified by the model as being at risk of heart attack.
- *True negatives (TN):* The number of individuals correctly identified by the model as not being at risk of heart attack.
- *False positives (FP):* The number of individuals incorrectly identified by the model as being at risk of heart attack.
- *False negatives (FN):* The number of individuals incorrectly identified by the model as not being at risk of heart attack.

Key metrics used

- *Precision:* TP/(TP + FP)
- *Recall:* TP/(TP + FN)
- *F1 Score:* 2*((Precision * Recall)/(Precision + Recall))
- *Support:* The number of occurrences of each class in the test dataset.
- *Accuracy:* A performance metric that measures the proportion of correctly predicted instances out of the total instances.

The following section will describe the Proposed ITARA method, which is a multi-criteria decision-making algorithm. The goal of using this method is to help prioritize individuals at risk of a heart attack so they can receive better services. Essentially, it suggests that the ITARA method will be applied to assess various factors (criteria) that affect the risk of heart attack, allowing for more informed decisions about which individuals need more immediate attention or resources.

Proposed ITARA method

The ITARA method, developed by Hatefi, is a widely recognized multi-criteria decision-making algorithm that assigns objective weights to risk factors based on their significance[34]. It relies on two key parameters: Dispersion Logic (DL) and the Indifference Threshold (IT). Dispersion Logic measures variability among ratings assigned to a risk factor; low variability reduces the factor's ability to distinguish between alternatives, leading to a lower weight. The Indifference Threshold sets the acceptable range of variability; when exceeded, the factor is considered significant because it can effectively differentiate between competing options. Together,

these parameters provide a systematic framework for identifying and ranking influential risk factors, thereby enhancing decision-making.

Despite its strengths, the original ITARA method has two notable limitations. First, it does not consider the distance of ratings from the aspiration level, defined as the best or ideal rating. This oversight can misrepresent the true importance of a risk factor. For example, consider two sets of ratings, (3, 1, 2, 2) and (9, 7, 8, 8), with an aspiration level of 1. Although both sets display similar variability, their distances from the aspiration level differ greatly. Yet, the original ITARA method assigns them equal weights, which contradicts practical reasoning[35]. Second, the ITARA method relies heavily on expert judgment to determine weights, which introduces subjectivity and potential bias into the analysis.

To overcome these limitations, we propose an enhancement to the ITARA method[36,37] specifically for prioritizing patients at risk of heart attacks. This improved approach integrates variable importance rankings derived from the random forest algorithm into the ITARA framework, particularly within the Indifference Threshold (IT). The random forest algorithm ranks variables based on their predictive relevance, ensuring that the most critical factors are emphasized during the analysis. Additionally, it replaces subjective expert judgment with objective, data-driven rankings, ensuring a transparent, consistent, and reproducible weighting process. This shift from subjective to objective weighting enhances the method's applicability in complex contexts, such as patient risk assessment and prioritization.

This section also aims to present a clear explanation of the proposed ITARA methodology, including detailed calculation steps for practical applications. The process can be summarized in three main steps:

1. Preparation

Like other MCDM methods, ITARA begins with a decision matrix. Assume there are m alternatives (M_i) evaluated across n risk factors (C_j) to obtain the assessment matrix $\overline{X}$, as shown in Eq. (1).

$$\overline{X} = [x_{ij}] = \begin{bmatrix} x_{11} & \cdots & x_{n,1} \\ \vdots & x_{ij} & \vdots \\ x_{m1} & \cdots & x_{m,n} \end{bmatrix}, i = 1, 2, \ldots, m; j = 1, 2, \ldots, n \tag{1}$$

Here, x_{ij} represents the risk rating of the $i-th$ alternative for the $j-th$ risk factor. Each risk factor is assigned an Indifference Threshold (IT) based on its characteristics. In this context, the IT for each risk factor is derived from its variable importance score (VI_j) obtained from the best-fit classification model, which quantifies each factor's contribution to the predictive model. Formally, this can be expressed as in Eq. (2):

$$IT_j = f\left(VI_j\right) \tag{2}$$

The function $f(\cdot)$ maps the feature importance scores to corresponding thresholds in ITARA, assigning greater weight to more influential risk factors in the classification model. By using this data-driven mapping, the framework replaces subjective expert judgment with an objective and reproducible procedure, enabling ITARA to prioritize patients based on quantifiable contributions of each variable to heart attack risk.

To allow fair comparison across factors with different scales or units, both the risk rating matrix x_{ij} and the IT_j vector are normalized, as shown in Eqs. (3) and (4), converting values to a unitless scale and ensuring consistent and meaningful ITARA calculations.

$$a_{ij} = \frac{x_{ij}}{\sum_i x_{ij}} \tag{3}$$

$$NIT_j = \frac{IT_j}{\sum_{i=1}^{m} x_{ij}}, j = 1, 2, \ldots, n \tag{4}$$

This approach effectively integrates data-driven importance scores into the multi-criteria evaluation, enhancing the objectivity and robustness of the risk assessment.

2. Calculating differences

To calculate the differences for each criterion, the ratings must first be sorted in ascending order. Denote $\overline{\beta}$ as the matrix is sorted by columns, so that $\beta_{ij} \leq \beta_{i+1,j}, \forall i \varepsilon \{1, \ldots, m-1\}$. Then, the matrix $\overline{\gamma}$ is created by computing the differences between consecutive elements of $\overline{\beta}$, as illustrated in Eq. (5).

$$\gamma_{ij} = \beta_{i+1,j} - \beta_{ij}, \forall i \varepsilon \{1, \ldots, m-1\} \tag{5}$$

3. Final weighting

Considerable differences can be calculated by comparing the $\overline{\gamma}$ elements by the indifference threshold and to calculate the $\overline{\delta}$ matrix as shown in Eq. (6):

$$\delta_{ij} = \begin{cases} \gamma_{ij} - NIT_j & For \gamma_{ij} > NIT_j \\ 0 & For \gamma_{ij} < NIT_j \end{cases} \tag{6}$$

Prediction	Reference	
	Less chance of heart attack	More chance of heart attack
Less chance of heart attack	169	11
More chance of heart attack	5	193

Table 2. Decision tree confusion matrix.

Prediction	Reference	
	Less chance of heart attack	More chance of heart attack
Less chance of heart attack	175	5
More chance of heart attack	2	196

Table 3. Random Forest Confusion Matrix.

Model	Heart attack possibility	Precision	Recall	F1-score	Support
Decision tree	No	0.97	0.94	0.95	180
	Yes	0.95	0.97	0.96	198
Random forest	No	0.99	0.97	0.98	180
	Yes	0.98	0.99	0.98	198

Table 4. Performance metric indices for classification algorithms evaluation.

Next, the root sum of squares $(\overline{v})$ is computed to aggregate the significant differences, which are then used to derive the final weights $(\overline{w})$ as presented in Eq. (7).

$$v_j = \sqrt{\sum_i \delta_{ij}^2} \Rightarrow w_j = \frac{v_j}{\sum_j v_j} \tag{7}$$

Result and discussion

This study focuses on identifying individuals at risk of heart attack using Kaggle datasets and the ITARA method, a multi-criteria decision-making algorithm. The ITARA framework is enhanced by removing its reliance on expert-assigned weights and integrating a classification algorithm to determine variable importance, which defines the Indifference Threshold (IT). Python was used to implement Random Forest and Decision tree models, selected for their high predictive performance in heart attack risk assessment and their ability to capture nonlinear relationships among multiple risk factors. The dataset was split into 80% training and 20% testing subsets.

A fivefold cross-validation was applied to the training data to ensure robust evaluation and reduce bias. Each model was then retrained on the full training set and evaluated on the independent test set, with performance metrics calculated to compare the models. Confusion matrices for the Decision tree and Random Forest algorithms are shown in Tables 2 and 3, respectively.

The confusion matrices in Tables 2 and 3 compare the classification outcomes of the Decision tree and Random Forest models. The Decision tree yielded 193 True Positives (TP), 169 True Negatives (TN), 11 False Positives (FP), and 5 False Negatives (FN), reflecting strong predictive ability but with a higher error rate. In contrast, the Random Forest achieved 196 TP, 175 TN, 5 FP, and 2 FN, demonstrating greater accuracy and reliability. As shown in Table 4, its higher precision, recall, and F1-score confirm a superior balance between sensitivity and specificity, making it more suitable for heart attack risk prediction.

The fivefold Cross-validation results further validate this conclusion. The Decision tree obtained a mean accuracy of 0.95 (SD = 0.008), while the Random Forest achieved 0.96 (SD = 0.01), indicating better stability and generalization. On the independent test set, the Decision tree reached 0.95 accuracy, whereas the Random Forest attained 0.97, confirming its robustness. Moreover, robust feature importance scores derived from the Random Forest were used as the Indifference Threshold (IT) in the ITARA method, enhancing its objectivity by replacing subjective expert weights with data-driven rankings.

Table 5 provides an overview of the feature importance scores, highlighting the variables that had the most significant impact on classification outcomes.

In the subsequent step, the ITARA method is applied by defining each patient in the dataset as an alternative, while the identified heart attack risk factors serve as the decision criteria for constructing the decision matrix. These criteria encompass clinical, demographic, and lifestyle variables such as age, cholesterol levels, blood pressure, and other relevant indicators that collectively shape an individual's risk profile. The detailed characteristics of these risk factors are presented in Table 6.

Factors	Importance
Age	0.079
Sex	0.033
Cp	0.136
Trestbps	0.075
Chol	0.081
Fbs	0.012
restecg	0.036
Thalach	0.105
Exang	0.034
Oldpeak	0.090
Slope	0.080
Ca	0.083
Thal	0.152

Table 5. Importance scores of features related to heart attack.

No	Criteria	Indifference threshold
1	Age	0.079
2	Sex	0.033
3	Cp	0.136
4	Trestbps	0.075
5	Chol	0.081
6	Fbs	0.012
7	restecg	0.036
8	Thalach	0.105
9	Exang	0.034
10	Oldpeak	0.090
11	Slope	0.080
12	Ca	0.083
13	Thal	0.152

Table 6. Heart attack risk criteria characteristics for the ITARA methodology.

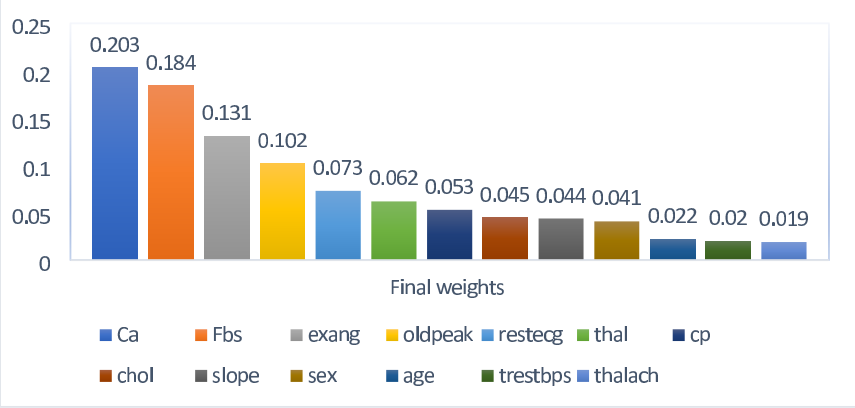


Fig. 3. Criteria weights using ITARA method.

After applying the ITARA method described earlier, each risk factor was assigned a weight reflecting its relative importance in predicting heart attack risk. Higher weights indicate a stronger influence in differentiating individuals across risk levels. These weights are used to prioritize patients, ensuring that those at the highest risk receive attention first. Figure 3 shows the distribution of the weights, providing a clear visualization of the relative significance of each factor.

The results shown in Fig. 3 highlight the number of major vessels (Ca) as the most influential risk factor for heart attacks, as it received the highest weight from the ITARA method. This finding underscores Ca's decisive role in prioritizing patients for timely medical attention and tailored interventions. The variable represents the number of major coronary arteries affected, with values ranging from 0 to 3. A value of 0 corresponds to no significant arterial blockage, suggesting minimal risk of coronary artery disease (CAD). In contrast, higher values reflect an increasing number of blocked arteries, signaling more severe disease progression and elevated heart attack risk. This scoring system provides a structured clinical measure of CAD severity, enabling physicians to stratify patients based on the urgency of care.

Managerial implications

Methodological part

This study makes a pivotal contribution to the field of MCDM by transforming the ITARA method into a fully data-driven and self-contained framework for weight assignment. Traditionally, MCDM approaches depend on expert judgments to determine criteria weights, which introduces subjectivity and potential bias. By integrating feature importance scores derived from machine learning models, this study eliminates the need for direct expert input in weight assignment. In this way, ITARA transitions into a fully data-driven method rather than one reliant on expert opinion, representing a significant advancement in MCDM methodology.

This shift is particularly valuable in contexts where expert reliance may lead to inconsistencies, biases, or where experts are not readily available. Deriving weights from machine learning ensures consistency, reproducibility, and scalability across diverse applications, including healthcare decision-making. For example, the objective ranking of heart attack risk factors enables decision-makers to prioritize resources and interventions with greater confidence.

It should be noted, however, that while ITARA no longer depends on expert input for weight assignment, experts remain essential in validating outcomes and ensuring alignment with real-world applications. This balance between data-driven objectivity and expert validation strengthens the method's practicality in high-stakes decision-making environments.

The introduction of this self-contained, data-driven framework also carries significant managerial implications. By providing an automated and objective tool for prioritization, ITARA enhances resource allocation and decision-making in healthcare and beyond. It paves the way for wider adoption of automated decision-support systems, reducing dependency on subjective expert judgment while maintaining the validity and reliability of results.

Patients at risk of heart attack

The findings of this study have several important managerial implications, particularly in healthcare decision-making. Applying the ITARA method to identify individuals at risk of heart attacks represents a significant advancement in prioritizing patients for timely interventions. By using feature importance scores derived from machine learning as criteria weights, the approach enables more accurate, data-driven assessments, allowing healthcare professionals to allocate resources more effectively to those most in need.

For healthcare managers, more precise prediction of heart attack risk supports improved decision-making in resource distribution, patient triage, and treatment prioritization. The data-driven framework reduces reliance on expert-based assessments, which can be subject to biases and inconsistencies, replacing them with objective, reproducible results. This ensures that the decision-making process is not only more efficient but also fairer and more consistent.

The use of the Random Forest algorithm to identify key risk factors, such as the number of major coronary arteries affected, further enhances the ability to focus on the most critical aspects of a patient's condition. This enables managers to make informed decisions about treatment strategies and prioritize high-risk individuals for immediate care.

Another notable managerial implication is scalability and adaptability. The ITARA method can be applied across diverse healthcare settings, from emergency departments to preventive care programs. Accurate identification of at-risk individuals can reduce healthcare costs, optimize resource utilization, and improve patient outcomes.

Additionally, the methodology's capacity to handle large datasets allows healthcare managers to continuously update and refine the model to reflect the latest research, changing patient demographics, and emerging risk factors. As healthcare increasingly embraces data-driven technologies, the implementation of such models can become a core component of predictive health management, supporting more personalized and proactive care strategies.

In conclusion, the enhanced ITARA method offers healthcare managers a powerful tool for enhancing patient outcomes. By adopting advanced, data-driven decision-making frameworks, healthcare systems can enhance efficiency, mitigate risks, and improve the quality of care for individuals at risk of heart attacks.

Conclusion

This study presents a novel adaptation of the ITARA method that integrates machine learning to improve the accuracy, reliability, and transparency of heart attack risk assessment. Traditional approaches often rely on expert-assigned weights, which can be subjective and inconsistent, potentially leading to variability in patient prioritization. In contrast, the proposed framework leverages feature importance scores from Random Forest, enabling fully data-driven and objective prioritization of patients at risk.

Patient medical histories, consisting of 13 variables including chest pain, blood pressure, blood sugar, oxygen saturation, and other vital indicators were used to train classification algorithms. The Random Forest classifier

achieved 97% accuracy, and its feature importance scores were directly used to define the Indifference Threshold (IT) in ITARA. Among the evaluated variables, the number of major coronary arteries (Ca) was identified as the most critical risk factor, providing an interpretable and actionable framework for guiding clinical decision-making.

Integrating machine learning with ITARA ensures that feature weighting is consistent, data-driven, and interpretable, reducing reliance on subjective expert judgment while maintaining a structured, multi-criteria approach. This hybrid framework enhances predictive reliability, allowing healthcare professionals to prioritize patients more effectively and support timely interventions and personalized treatment strategies.

Despite these strengths, some limitations remain, including potential dataset biases, lack of clinical validation, and the risk of overfitting. To address these issues, future research could incorporate independent clinical datasets to enhance external validity and assess the generalizability of the findings across diverse populations. Overall, the RF–ITARA framework demonstrates strong potential to enhance healthcare decision-making by minimizing subjectivity, improving predictive accuracy, and facilitating efficient, individualized patient interventions.

Future work

Future studies should evaluate the RF–ITARA framework across diverse populations and clinical settings to ensure generalizability. Validation using multicenter datasets and real-world hospital data would strengthen the applicability of the framework in different healthcare contexts.

Integration with other AI models represents a promising direction, particularly with methods that provide reliable and interpretable feature importance, such as explainable deep learning or reinforcement learning models. Techniques such as SHAP (SHapley Additive exPlanations) or permutation-based feature importance could enable ITARA to leverage these advanced models, thereby supporting adaptive, real-time decision-making.

Using longitudinal patient data could allow the framework to capture dynamic risk factors over time, enhancing prediction of both immediate and long-term risk. Hybrid approaches combining ITARA with electronic health records, wearable devices, and other digital health data may facilitate more proactive and personalized interventions. Comparative studies against other multi-criteria decision-making and machine learning approaches would further clarify ITARA's strengths and limitations, ultimately supporting more effective management of heart attack risk.

Data availability

All data generated or analyzed during this study are included in this published article.

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Author contributions

N.A., S.H.Z and F.C. worked together to accomplish this work, N.A. and S.H.Z. wrote the original draft and worked on methods and calculations, F.C. supervised the paper.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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